

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Phase 3: System & Technical Design Architecture Specification

LEAD ARCHITECT	DOCUMENT STATUS	TARGET RELEASE	DATE
ServiceNow Platform Team	Design Freeze	Vancouver Patch 4	Q3 2026

1. SYSTEM ARCHITECTURE OVERVIEW

The System Design Phase converts functional requirements into an integrated architecture within ServiceNow. The platform architecture connects the Service Catalog user interface, decision routing engines, integration spokes, and Hardware Asset Management (HAM) tables.



2. SERVICE CATALOG ITEM SPECIFICATIONS

The catalog item `cat_item_standard_laptop` provides a dynamic UI that adapts to user roles and location policies.

Variable Name	Variable Type	Default / Dynamic Value	Validation & UI Logic
<code>requested_for</code>	Reference (sys_user)	<code>gs.getUserID()</code>	Auto-populates manager, department, and cost center.
<code>hardware_tier</code>	Select Box	Standard Business (Tier 1)	Options: Tier 1 (Standard), Tier 2 (Developer), Tier 3 (Exec).
<code>fulfillment_mode</code>	Lookup Select	Dynamic Scripted Choice	Evaluates local stock inventory before rendering options.
<code>shipping_address</code>	Custom / Text	<code>requested_for.location</code>	Editable for remote employees; validated against postal API.

3. UI POLICIES & CLIENT-SIDE SCRIPTING STRATEGY

Catalog UI Policy: Developer Bundle Justification Condition: <code>hardware_tier == 'Tier 2 (Developer)'</code> Action: Set variable <code>business_justification</code> to <code>Mandatory = True</code> and <code>Visible = True</code>. Enforces budget justification prior to submit.	Catalog Client Script: User Location Auto-Fill Type: <code>onChange (requested_for)</code> Logic: Executes <code>GlideAjax</code> call to fetch user record attributes (Cost Center, Office Address, Manager) to eliminate manual entry.
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4. FLOW DESIGNER WORKFLOW ARCHITECTURE

The core execution logic is built within ServiceNow Flow Designer using a structured master flow with dedicated sub-flows for execution safety and reusability.

4.1 Main Flow Execution Logic (flow_laptop_procurement)

- Trigger:** Service Catalog record created in sc_req_item where Item IS cat_item_standard_laptop.
- Stage 1 - Get Catalog Variables:** Extract hardware_tier, requested_for, and shipping_address.
- Stage 2 - Call Sub-flow:** Execute subflow_procurement_approval passing Request Item record and Tier level.
- Stage 3 - Conditional Branch:**
 - If Approved:* Advance to Inventory Allocation.
 - If Rejected:* Update RITM state to Closed Rejected and notify user via MS Teams.
- Stage 4 - Call Sub-flow:** Execute subflow_fulfill_and_tag_asset for stock reserve or REST vendor dispatch.

5. DECISION TABLE SPECIFICATION: DYNAMIC APPROVAL MATRIX

Instead of hardcoded script logic, dynamic decision tables route approval workflows based on budget limits and hardware tiers.

Decision Name	Input Condition 1 (Tier)	Input Condition 2 (Cost)	Output Routing Result
DT_Rule_01	Tier 1 (Standard)	≤ \$1,500 USD	Auto-Approve (Skip Manager)
DT_Rule_02	Tier 2 (Developer)	> \$1,500 AND ≤ \$2,800 USD	Require Direct Manager Approval
DT_Rule_03	Tier 3 (Executive)	> \$2,800 USD	Require Manager + VP Finance Approval

6. VENDOR REST SPOKE & INTEGRATION TECHNICAL DESIGN

When stock is unavailable in local inventory, Flow Designer triggers an outbound REST call to the primary hardware supplier API.

Outbound REST Spoke Configuration Specifications

Endpoint: `https://api.vendor-hardware.com/v2/orders/purchase`

Authentication: OAuth 2.0 Client Credentials Grant (Stored in ServiceNow Credentials Table discovery_credentials)

Payload Schema (JSON):

```
{
  "purchaseOrder": "PO-SERVICENOW-RITM0010928",
  "itemSKU": "LAPTOP-DEV-16RAM-1TB",
  "recipient": { "name": "John Doe", "address": "123 Tech Way, San Jose CA" }
}
```

7. CMDB DATA MODEL & ASSET SCHEMA DESIGN

To maintain accurate configuration state, automated script actions map catalog outputs directly into Hardware Asset Management (HAM) and CMDB CI tables.

Target Table	Field Name	Source Attribute	State Transition
alm_hardware	asset_tag	Auto-Generated Sequence	In Stock → Reserved
alm_hardware	assigned_to	sc_req_item.requested_for	Assigned
cmdb_ci_computer	serial_number	REST Payload / Vendor Webhook	Pending Allocation → In Use
cmdb_ci_computer	mac_address	Staging Task / Auto-Discovery	Operational

8. SECURITY, ROLES & ACCESS CONTROLS (ACLS)

Role-Based Access Control (RBAC) - sn_procurement.user: Access to order standard laptops. - sn_procurement.admin: Ability to modify Decision Tables and API configs. - asset: Read/Write access to inventory updates.	Data Protection & Logging Safety - Restricted log levels on REST Spoke inputs to avoid writing shipping address PII into system log files. - Encrypted HTTP header transmission for API credentials.
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9. DESIGN SIGN-OFF & BUILD SPRINT SCHEDULE

Phase 3 Gate Review Status: APPROVED FOR BUILD Technical design baseline is completed and frozen. Build execution will proceed according to the sprint schedule below:
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Sprint	Deliverable Target	Engineering Focus	Status
Sprint 1	Catalog Form & UI Policies	Service Catalog items, dynamic variables, Client Scripts.	Scheduled
Sprint 2	Flow Designer & Decisions	Main flow, sub-flows, Decision Tables, MS Teams integration.	Scheduled
Sprint 3	REST Spoke & CMDB Sync	Vendor API outbound integration & Asset state mapping.	Scheduled
Sprint 4	UAT & Deployment	End-to-end testing, error handling, production release.	Scheduled